
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has become a central tool for improving the reasoning
2 capabilities of language models, yet the training dynamics that produce these
3 gains remain insufficiently explained. This paper makes the contribution structure
4 explicit around one claim: RL improves reasoning primarily by changing which
5 existing reasoning patterns a model selects, rather than by uniformly improving
6 every pattern. Empirically, we combine reasoning-pattern-level and token-level
7 analyses across RL training and show that individual reasoning patterns maintain
8 relatively stable success rates while RL concentrates probability mass through a
9 sparse subset of critical tokens, thereby reshaping the distribution over patterns
10 that drives performance. Theoretically, we introduce a question–reason–answer
11 framework for analyzing this selection process under two representative reward
12 mechanisms: verifiable reward (RLVR) and model internal feedback (RLIF). For
13 RLVR, our analysis identifies two regimes—rapid convergence to high-success
14 reasoning strategies and a harder entanglement phase—and shows why the base
15 model’s initial reasoning quality is decisive for convergence. For RLIF, we explain
16 both the early performance gains induced by internal rewards and the possible
17 degradation that can emerge with continued training. Experiments, case studies, and
18 numerical simulations validate these empirical and theoretical findings, yielding a
19 reviewer-facing account of how RL reshapes reasoning in LLM post-training.

20 1 Introduction

21 State-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14] now
22 achieve strong performance on complex logical tasks, including mathematics [30, 44, 41] and
23 programming [49, 40, 8]. A central contribution of this paper is to clarify what reinforcement learning
24 (RL) contributes to this success: we study how RL changes the model’s selection among reasoning
25 patterns during post-training.

26 Prior work has begun to probe this mechanism, but the training dynamics remain only partially
27 characterized. By comparing pass@k performance, Yue et al. [42] show that RL post-trained models
28 often struggle to surpass base models, suggesting that RL may not create fundamentally new reasoning
29 patterns. Entropy-based analyses show that RL methods can reduce policy entropy [9, 45], and Cui
30 et al. [9] further connects entropy changes to model performance. Complementary token-level studies
31 find that RL mainly adjusts a sparse set of critical tokens [33, 17]. What is still missing, and what
32 this paper contributes, is an empirical and theoretical account that links these token-level shifts to
33 reasoning-pattern selection across the RL trajectory.

34 Our first contribution is an empirical decomposition of RL training dynamics at two complementary
35 resolutions. At the token level, we examine how next-token prediction rankings shift during training.
36 At the reasoning-pattern level, we use LLM-based and rule-based procedures to extract, classify, and
37 track response patterns together with their success rates and distributions. This analysis sharpens the

evidence from prior work [17, 42]: RL primarily optimizes a sparse subset of critical tokens, and those token changes *reshape reasoning pattern distributions* that determine model performance. At the same time, the intrinsic success rate of each individual pattern remains relatively stable, motivating a theory centered on pattern selection rather than uniform pattern improvement.

Our second contribution is a theoretical framework that makes this selection view precise. We model reasoning as a two-stage question–reason–answer process $q \rightarrow r \rightarrow a$: first the model selects a reasoning pattern according to $\pi(r|q)$, and then it generates an answer according to $\pi(a|r, q)$. Within this framework, we analyze two representative reward types: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we show convergence toward the highest-success reasoning pattern r^* and identify two regimes: rapid convergence when the base model already has strong reasoning quality, and entanglement-stage optimization challenges when it does not. For RLIF, we explain why internal rewards can initially improve performance while also showing how continued training may converge to a state worse than the base model. Additional experiments then serve as a third contribution by validating the theory through case studies and simulations.

The paper’s contributions are therefore organized as follows:

- **Empirical contribution:** We conduct a systematic token-level and reasoning-pattern-level analysis across RL training. By combining next-token ranking shifts with LLM-based and rule-based pattern analysis, we show more directly than prior work [17, 42] that RL optimizes a sparse subset of critical tokens, *reshapes reasoning pattern distributions*, and leaves individual pattern success rates relatively stable.
- **Theoretical contribution:** We introduce a two-stage $q \rightarrow r \rightarrow a$ framework that analyzes RL as reasoning-pattern selection. The framework proves RLVR convergence toward the highest-success pattern, separates rapid-convergence and entanglement regimes, and explains both the early gains and possible later degradation of RLIF.
- **Validation contribution:** We support the theory with additional case studies and numerical simulations. These results connect the abstract dynamics to observed LLM post-training behavior and make the practical implications of reasoning-pattern selection explicit.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning has become a major post-training mechanism for large language models (LLMs), with demonstrated value for human-preference alignment [25, 48] and for mathematical and programming reasoning [30, 20, 22]. This literature motivates our contribution but also clarifies its boundary: we do not propose a new RL algorithm or reward source; instead, we analyze how existing RL objectives change reasoning behavior. Classical RL pipelines rely on reward models trained from human-annotated data [26, 25, 2], whereas recent reasoning systems increasingly use more accessible signals such as reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. Algorithmic advances such as GRPO [30], DAPO [41], and VAPO [43] further highlight the need for a mechanism-level account of why these training procedures reshape reasoning.

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. Mechanistic and theoretical studies of RL for LLMs provide the closest context for our paper. Early work emphasized high-level questions such as reward design [28, 18, 38] and training-algorithm development [37, 11, 21]. As RLVR and RLIF simplified the reward interface, more recent work has asked why such signals work at all: Shao et al. [29] argue that spurious rewards can unlock latent pretraining skills, Agarwal et al. [4] relate performance gains to entropy minimization, and Gandhi et al. [13] identify cognitive behaviors such as backtracking that correlate with stronger reasoning. Related analyses show that RL fine-tuning amplifies pre-trained behaviors and drives convergence toward dominant output formats [46], a phenomenon also studied as entropy collapse by Wang et al. [34], Cui et al. [10]. Our contribution is complementary and more targeted: we make reasoning-pattern selection the central object of analysis, connect it to sparse token-level changes, and provide a mathematical framework for RLVR and RLIF training dynamics.

89 3 Preliminaries

90 **Reinforcement Learning for LLMs** This section fixes the RL notation used by both our empirical
 91 framing and theoretical contribution. Let π_θ be a language model with parameters θ , treated as the
 92 policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ generates an
 93 answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. We write the general KL-regularized RL objective as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

94 Here π_{ref} denotes the base reference policy, and β controls the KL penalty that keeps the optimized
 95 policy from drifting excessively away from π_{ref} . This regularized objective is the common substrate
 96 on which we compare the contribution of different reward signals to reasoning-pattern selection.

97 The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ can take several forms, including a trained reward model [25] or a
 98 rule-based scoring function [30]. To keep the analysis focused and reviewer-visible, this paper studies
 99 two representative reward mechanisms: verifiable reward (RLVR) [14, 30] and the model’s internal
 100 feedback (RLIF) [47, 4].

101 For RLVR, the contribution-relevant property is that the reward directly tests whether the generated
 102 answer $\mathbf{y}$ matches the correct answer for question $\mathbf{x}$. A typical binary reward is:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

103 RLIF [47, 4] instead uses an intrinsic signal derived from the model itself, making it a useful contrast
 104 for our training-dynamics analysis. Following Zhao et al. [47], we instantiate the RLIF reward as
 105 the negative average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the
 106 model’s next-token distribution:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \| \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

107 With these definitions in place, the rest of the paper asks how RLVR and RLIF alter the distribution
 108 over reasoning patterns, and why those alterations can produce either stable gains or later degradation.

109 4 Experimental Exploration for RL Training

110 This section front-loads the paper’s empirical contribution: we show, across reward types and analysis
 111 scales, that RL changes reasoning performance mainly by reallocating probability among existing
 112 reasoning patterns. The evidence is organized in three increasingly fine-grained steps. First, matched
 113 RLVR and RLIF training trajectories show that verifiable rewards produce steady gains, whereas
 114 internal-feedback rewards can become unstable and degrade performance. Second, reasoning-pattern
 115 analysis explains this contrast: RLVR increases the use of high-success patterns, while RLIF does
 116 not reliably specialize in them. Third, token-level analysis identifies a sparse mechanism capable of
 117 producing those distributional shifts, because only a small fraction of tokens undergo probability-rank
 118 changes. Relative to prior analyses [17, 42], the contribution of this section is therefore a unified
 119 empirical account: RL optimizes sparse critical tokens and thereby *reshapes reasoning pattern*
 120 *distributions*.

121 4.1 Training Procedure Overview for RLVR and RLIF

122 We begin with the highest-level evidence for the empirical contribution: when the model and data are
 123 fixed, the reward signal alone induces qualitatively different reasoning-training dynamics. For this
 124 controlled comparison, we use Qwen2.5-3B [39] as the base model, train on the MATH dataset [16],
 125 and keep all non-reward settings identical between RLVR and RLIF.

126 Figure 1a provides the first result in this contribution chain. RLVR yields a stable trajectory in which
 127 performance improves and gradually converges, whereas RLIF improves performance early but can
 128 later decrease it, sometimes below the original base model. This contrast motivates the remaining
 129 analyses: to explain why matched training runs diverge, we must identify how each reward changes
 130 the model’s distribution over reasoning behavior.

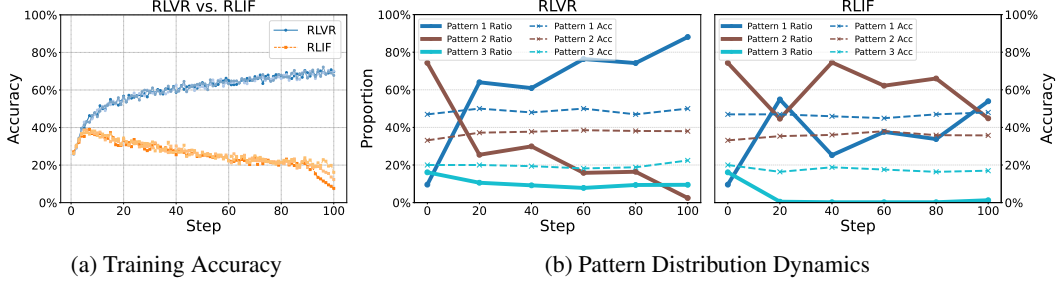


Figure 1: Empirical evidence for the selection-based contribution. **(a)** Training trajectories for RLVR and RLIF on the MATH dataset: RLVR gives stable gains, whereas RLIF first improves and then decreases model performance; each training paradigm is repeated with different random seeds. **(b)** Reasoning-pattern comparison: solid lines show each pattern’s response proportion (left axis), and dotted lines show that pattern’s accuracy (right axis). RLVR gradually increases high-accuracy patterns and suppresses lower-accuracy patterns, while RLIF shows unstable pattern dynamics. For readability, patterns are sorted and named from high to low according to their corresponding accuracy.

4.2 Reasoning-pattern Level Analysis

The second empirical contribution translates the performance contrast into reasoning-pattern dynamics. We collect base-model responses and use GPT-4o [19] to group them into categories according to keywords and logical structure, assigning a description to each pattern. By fixing these categories before tracking training, we can separately measure two quantities that are often conflated: how frequently each reasoning pattern is selected and how successful that pattern is when selected.

Figure 1b makes this contribution concrete on Number Theory tasks from MATH [16], with additional models and datasets reported in Appendix A. The pattern-level analysis gives three reviewer-facing findings that support the selection-based thesis:

- RLVR-trained models consistently shift toward higher-success reasoning patterns, explaining why overall accuracy improves steadily under RLVR.
- RLIF produces unstable pattern distributions and does not reliably specialize in the more effective patterns.
- For both methods, the success rate of an individual reasoning pattern remains stable across training.

Together, these findings locate the empirical mechanism at the level of pattern selection. RLVR improves performance mainly by changing which patterns are selected, not by substantially improving the intrinsic success rate of each pattern. The remaining question is how large a token-level change is needed to induce this visible redistribution over reasoning patterns.

4.3 Token-level Analysis

The third empirical contribution connects pattern selection to sparse token changes. Inspired by Huan et al. [17], Yue et al. [42], we sample base-model responses for questions from GSM8K [6], MATH [16], and AIME [7]. For each identical response, we compare token ranks under the base model and the RL-enhanced model, where a token’s rank is the position of its probability among all vocabulary tokens given the preceding context. Table 1 shows that token ranks remain largely stable: in most cases, fewer than 10% of tokens experience a rank shift. The contribution-level implication is that large changes in pattern usage do not require wholesale rewriting of reasoning traces; selective probability changes on sparse critical tokens can redirect the model toward different reasoning patterns.

Table 1: Token ranking-change ratios after RL. The table supports the sparse-change component of the empirical contribution: only a small subset of tokens changes rank across training.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

164 Across these three empirical views, the section’s contribution can be summarized as follows: *RL*
 165 *primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distribu-*
 166 *tions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains*
 167 *relatively stable during training.*

168 These empirical findings set up the theoretical contribution. If RL mainly changes sparse token
 169 choices and thereby redistributes probability across stable-success reasoning patterns, then the
 170 appropriate theoretical object is not only the final answer accuracy but also the training dynamics of
 171 reasoning-pattern selection.

172 5 Theoretical Characterization of RL Training Dynamics

173 This section provides the paper’s theoretical contribution: a formal explanation of why the empirical
 174 pattern-selection view can account for RL training dynamics. We first abstract reasoning into two
 175 stages—selecting a reasoning pattern and then deriving an answer—so that RL’s main effect can
 176 be analyzed as changes to the first stage. Within this framework, we characterize how two reward
 177 types behave. For RLVR, we identify two convergence regimes that explain when training efficiently
 178 selects the best pattern and when it enters an entanglement phase. For RLIF, we show why internal
 179 rewards can improve accuracy at initialization while still permitting convergence to states worse than
 180 the base model.

181 5.1 A Theoretical Framework for Reasoning Models

182 **Abstract Reasoning Process.** The framework isolates the object that our experiments identify
 183 as central: the distribution over reasoning patterns. Given a question q , the model first selects a
 184 reasoning pattern r from candidate patterns $\mathcal{R} = \{r_1, r_2, \dots\}$ and then generates a final answer
 185 $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ accordingly. The model selects pattern r_i with probability $p(r_i|q)$, and each
 186 pattern has its own success rate. Under this abstraction, the RLIF optimization objective (Eq. 3.3) can
 187 be rewritten as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

188 **Policy Parameterization.** To connect the abstraction to language-model policies, let $\mathcal{V} = \{q\} \cup$
 189 $\mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and let $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ represent the output sequences. Given a
 190 question q , the language model produces a distribution over output sequences $y \in \mathcal{Y}$ autoregressively:

$$\text{(general policy)} \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

191 Here $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ , and the model predicts the l -th token from the previous
 192 context $y_{<l}$. The next-token distribution is $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ for $z =$
 193 $f_\theta(x, y_{<l})$.

194 Because practical LLMs make direct optimization guarantees difficult [3, 23], our theoretical contri-
 195 bution follows prior work [27, 24, 10, 45] and analyzes a *tabular policy*. This parameterization is a
 196 special case of Eq. 5.2: each output receives a trainable logit for the corresponding previous token, so
 197 for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

198 The column $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ corresponds to the previous token y_{l-1} .

199 **Optimization for RL.** We analyze training dynamics by combining the tabular policy in Eq.5.3
 200 with the RL objective $\phi_{\text{RL}}(\theta)$ in Eq.3.1. In the small-learning-rate limit, the policy-gradient dynamics
 201 are represented by gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

Here $\theta(t)$ denotes the parameters of $\pi_{\theta(t)}$ at training time t , initialized as $\theta(0) = \theta_{\text{ref}}$.

The key empirical observation to encode is that individual pattern success rates remain stable during training. We interpret this as a separation between two optimization difficulties: changing the mapping from questions to reasoning patterns is easier than changing the mapping from a selected reasoning pattern to the final answer. We formalize this contribution-critical observation as follows:

Assumption 5.1. *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer $\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

For the rest of the section, we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$ for the fixed success rate of pattern $\mathbf{r}$. We first derive the optimal policy for the RL objective (Eq.3.1) under the general autoregressive policy (Eq.5.2), and then use the tabular policy (Eq. 5.3) to analyze RLVR and RLIF training dynamics.

5.2 The Optimal Policy for RL

We first state the optimal-policy result that formalizes the empirical contrast between RLVR and RLIF. The contribution of this proposition is to show that RLVR rewards success rates directly, whereas RLIF rewards confidence properties that need not identify the correct answer:

Proposition 5.2. *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 makes the selection mechanism explicit. Assuming π_{opt} lies in the general autoregressive policy class, which includes both LLMs and the tabular policies used below, RLVR increases the probability of a reasoning pattern $\mathbf{r}$ when its success rate $p^*(\mathbf{r})$ is large enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. RLIF behaves differently: $R_{\text{RLIF}}(\mathbf{r})$ depends only on confidence in the final-answer distribution and therefore does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. Thus, the optimal RLVR policy is aligned with higher reasoning-pattern success, while the optimal RLIF policy does not by itself guarantee accuracy improvements over the base model.

The limiting case also clarifies why pattern selection is the right theoretical lens. Since β is typically small in practice (e.g., 0.001) [26, 5, 44], as $\beta \rightarrow 0$ we obtain:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit, the policy becomes a deterministic selector of the reasoning pattern $\mathbf{r}$ with the largest $R(\mathbf{r})$, regardless of the reference model’s initial pattern distribution. This optimal-policy view explains the target of training, but not the path taken to reach it. Because the optimization landscape is non-convex, we next analyze tabular-policy dynamics (Eq. 5.3) to explain when RLVR reaches high-reward patterns efficiently and why RLIF can show early gains before later degradation.

5.3 Training Dynamic Analysis for RLVR

The RLVR analysis contributes a convergence explanation for the empirical observation that base-model quality matters. We focus on the *optimal reasoning pattern* $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$, the pattern with the highest success rate for reaching the correct answer $\mathbf{a}^*$. Gradient-flow dynamics reveal two regimes. In the first, the probability $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ of the optimal pattern increases steadily until it converges to 1. In the second, the model enters an entanglement stage in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ slows optimization of the optimal pattern; after this stage, training transitions to the first-regime dynamics and converges to $\mathbf{r}^*$.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$, satisfies:*

$$(Regime 1) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 states the positive part of the RLVR contribution. When the reference model is already strong enough that its overall accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7), RLVR efficiently drives selection toward the optimal pattern at rate $\mathcal{O}(1/\epsilon)$ and achieves overall accuracy close to the success rate of that optimal pattern.

The second regime explains why RLVR can nevertheless be difficult in practice [44, 36], especially when the base reference model is weaker. We analyze the case where the model first passes through an entanglement stage caused by the *suboptimal reasoning pattern with the second-highest success rate*, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, which slows the shift toward the optimal reasoning pattern:

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(Regime 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that the model is guaranteed to transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

Theorem 5.4 studies a special regime in which only the optimal and second-best patterns have success rates above the model’s overall accuracy. We use this setting for analytical clarity; the same selection-based phenomenon should extend to more general cases where *at least* two reasoning patterns exceed the average, though the analysis would be more involved.

The critical contribution of Theorem 5.4 is the time-scale separation. RLVR may need T_0 steps before the overall success rate exceeds that of the suboptimal pattern, thereby reaching the favorable regime in Theorem 5.3 where the optimal pattern has sufficiently large selection probability. Although Theorem 5.3 gives a convergence time T_1 polynomial in $1/\epsilon$, the T_0 term in Theorem 5.4 may grow super-exponentially with $\gamma_{\pi_{\text{ref}}}$, the ratio between the total success rate through $\mathcal{R}/\{\mathbf{r}'\}$ and the success rate through $\mathbf{r}^*$. When the base model assigns very small probability to the optimal pattern, $\gamma_{\pi_{\text{ref}}}$ can be large and T_0 can become prohibitively long. We refer to this prolonged period before reliable optimal-pattern selection as the entanglement stage.

Sec 6 provides case studies and simulations that make these two theoretical regimes visible empirically.

5.4 Theoretical Explanation for RLIF

The RLIF analysis contributes an explanation for the empirical rise-and-fall behavior observed earlier. Proposition 5.2 shows that RLIF, by rewarding confidence in the final-answer distribution, does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. This creates a path to performance degradation relative to the base model. At the same time, prior work shows that RLIF can improve performance without external rewards Agarwal et al. [4], Zhao et al. [47]; we therefore analyze why a well-trained LLM can improve at the initial stage even under this imperfect signal.

We start from a well-trained LLM satisfying the following majority condition:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r})\pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

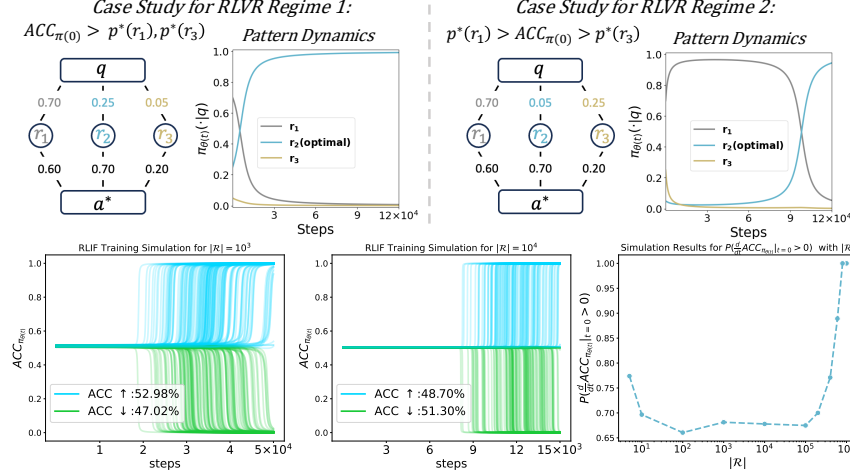


Figure 2: Validation evidence for the theoretical contribution. The RLVR case studies (upper) visualize the two predicted regimes: rapid convergence (left) and the entanglement phase (right). The RLIF simulations (bottom) show convergence probabilities to different states (left) and the probability of initial performance gains (right) as $|\mathcal{R}|$ varies.

Because majority voting [35] is effective for improving model performance, modern well-trained LLMs often assign higher aggregate probability to the correct answer $\mathbf{a}^*$ than to incorrect alternatives, making Assumption 5.5 plausible. Under this condition, we analyze the simplified case $|\mathcal{A}| = 2$ in which each reasoning path’s success rate follows a uniform distribution and pattern selection is high-entropy². This setting isolates the initial RLIF accuracy dynamics and yields the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

Theorem 5.6 explains both sides of RLIF’s empirical behavior. Its first statement shows that accuracy increases at initialization, accounting for RLIF’s early gains. Theorem 5.6’s second statement, combined with Proposition 5.2, explains the later risk: when $\beta = 0$, training converges deterministically to the pattern maximizing $R_{\text{RLIF}}(\mathbf{r})$, and with probability 0.5 this pattern is also the least successful one, $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. Thus RLIF can initially improve performance because the accuracy derivative is positive, yet continued training can favor low-accuracy reasoning paths and ultimately fall below the base model.

6 Case Studies and Numerical Simulations

This section provides the validation contribution that links the theoretical results back to observable training behavior. For RLVR, the case studies in Fig. 2 (upper) instantiate the two regimes from Theorems 5.3 and 5.4. For RLIF, the simulations in Fig. 2 (bottom) sample reasoning-path success rates from a uniform distribution and test both parts of Theorem 5.6: early-step improvement and eventual convergence behavior.

6.1 Case Studies for RLVR

Section 5.3 predicts two RLVR regimes, and the purpose of these case studies is to make their practical distinction visible. In the first regime (Eq 5.7), the probability of selecting the optimal

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

reasoning pattern $\pi_{\theta}(\mathbf{r}^*|q)$ increases monotonically until convergence. In the second regime (Eq 5.8), an initial entanglement phase delays the transition to the first-regime dynamics and therefore delays convergence to $\mathbf{r}^*$. We examine both cases:

- Fig. 2 (upper left) validates the rapid-convergence regime: when the reference model satisfies Eq 5.7, so that its overall accuracy exceeds the success rates of all non-optimal patterns, training quickly concentrates on the optimal reasoning pattern.
- Fig. 2 (upper right) validates the entanglement regime from Theorem 5.4. When a suboptimal reasoning pattern $\mathbf{r}'$ has higher success rate than the initial overall accuracy, training must pass through the transition period T_0 defined in Theorem 5.4 before entering rapid optimization. In the shown scenario, $\gamma_{\pi_{\text{ref}}}$ from Eq 5.9 is large ($\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$), so T_0 scales with $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$ and becomes prohibitively long. The resulting pattern dynamics in Fig. 2 display the predicted entanglement stage and align with the theorem.

6.2 Numerical Simulations For RLIF

Section 5.4 predicts a distinctive RLIF signature: under Assumption 5.5 and the base-model distribution in Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, yet convergence to the least accurate reasoning path occurs with probability 50%. The simulations validate both claims. Fig. 2 (bottom left) shows that, across varying $|\mathcal{R}|$, the probability of converging to an improved state ($ACC \uparrow$) is approximately equal to the probability of converging to a degraded state ($ACC \downarrow$), supporting Theorem 5.6’s second statement. Fig. 2 (bottom right) shows that for large $|\mathcal{R}|$, the probability of initial performance improvement approaches 1, supporting the theorem’s first statement.

7 Conclusions and Limitations

This paper contributes a selection-based account of how reinforcement learning reshapes reasoning in LLMs. Empirically, we show that RL training changes performance mainly by shifting probability mass across reasoning patterns: individual pattern success rates remain relatively stable, while sparse critical-token changes alter which patterns are selected. Theoretically, we formalize this observation with a question–reason–answer framework and analyze two representative reward mechanisms. For RLVR, the analysis identifies both rapid convergence to high-success reasoning patterns and a slower entanglement regime that depends on base-model quality. For RLIF, the analysis explains early accuracy gains while also showing why continued training can converge to worse-than-base behavior. Case studies and simulations validate these claims and make the contribution structure visible across empirical, theoretical, and validation evidence.

The work also has limitations that define useful next steps. The interpretability and reliability of LLM-identified reasoning patterns require additional validation. The theoretical framework abstracts real reasoning into simplified pattern-selection dynamics and should be extended to more complex real-world reasoning settings. Finally, the current analysis relies on specific base-model assumptions, and broader generalization across architectures and training conditions remains an important direction for future work.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

This appendix extends the empirical contribution from Section 4.2 beyond the Number Theory tasks used in the main text. We additionally analyze Geometry tasks, shown in Figure 3, to test whether the same pattern-selection dynamics appear in another mathematical domain. The result supports the main claim: under RLVR, the model increasingly selects higher-accuracy reasoning patterns, whereas RLIF produces unstable pattern selection. This supplementary evidence aligns with both the empirical observations and the theoretical regimes described in the main paper.

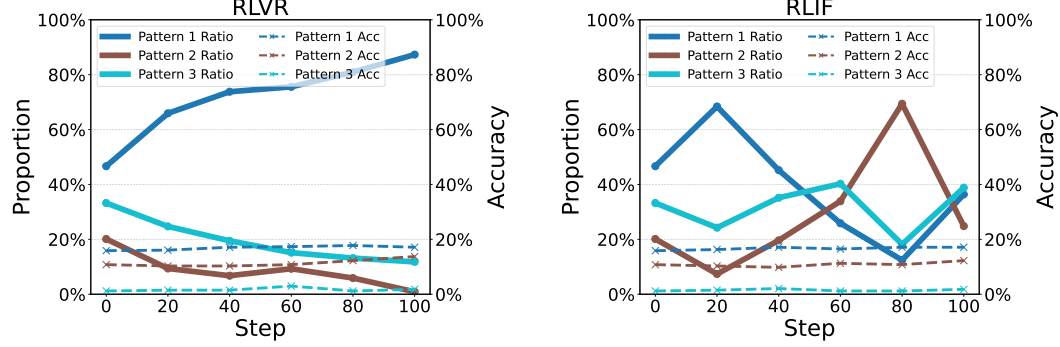


Figure 3: Geometry-task pattern dynamics for RL-enhanced models. RLVR concentrates probability on the highest-accuracy pattern (Pattern 1), while RLIF shows unstable selection behavior, reinforcing the paper’s pattern-selection explanation.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether the paper’s findings generalize across models, we compare Qwen-2.5-7B-Instruct [39] with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source RLVR-enhanced model for mathematical reasoning. Figure 4 reports supplementary tasks from number theory, geometry, algebra, calculus, counting and probability, and AMC23, using challenging problems from MATH [16] and AMC23 [1]. Across these domains, the same contribution-level pattern appears: RLVR enhancement increases the frequency of high-accuracy reasoning patterns and reduces the use of less effective patterns. Appendix B.3 provides task-specific details.

The cross-task pattern analysis reinforces two parts of the paper’s contribution. First, the highest-success patterns consistently become more prevalent after RLVR enhancement, supporting the pattern-selection explanation from Section 4 and Section 5. Second, tasks such as Algebra and AMC23-19 in Figure 4 show stable success rates for individual reasoning patterns, providing additional empirical support for Assumption 5.1.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

We further test the theory on a larger RLVR-enhanced model, QwQ-32B [32], which is based on Qwen-2.5-32B-Instruct [39]. The goal of this appendix experiment is to check whether the pattern-selection account remains visible at larger scale. We compare reasoning-pattern shifts on four tasks: number theory and geometry from MATH [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The results show that most QwQ-32B reasoning patterns correspond to the most accurate patterns identified for Qwen-2.5-32B, which is consistent with the theoretical prediction that RLVR concentrates probability on high-success patterns. We do not report QwQ pattern accuracy because the model is accessed through an API, and we cannot extract the CoT data needed for the base model to continue generation. Consequently, we cannot measure reasoning-pattern accuracy separately from model-level accuracy. Appendix B.2 explains this limitation under “Accuracy Analysis”.

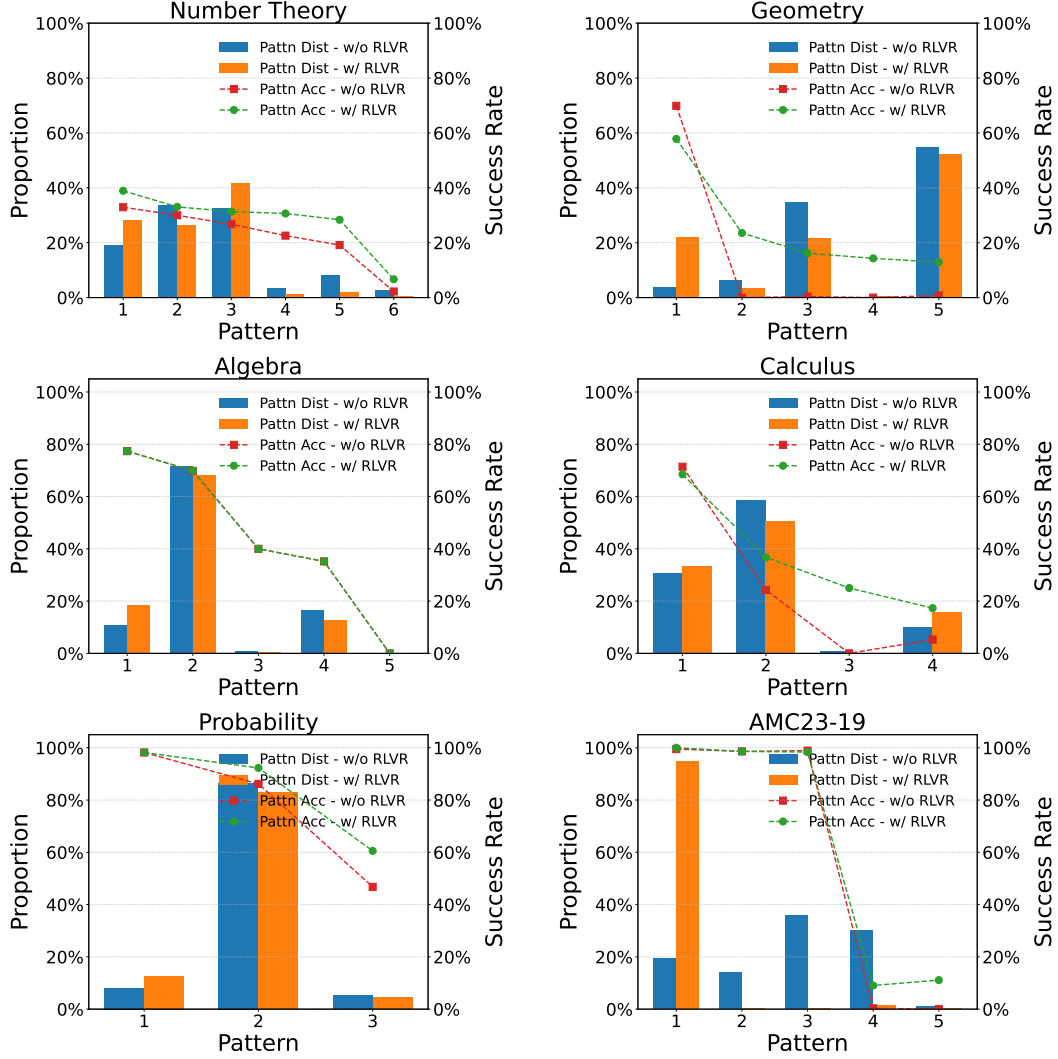
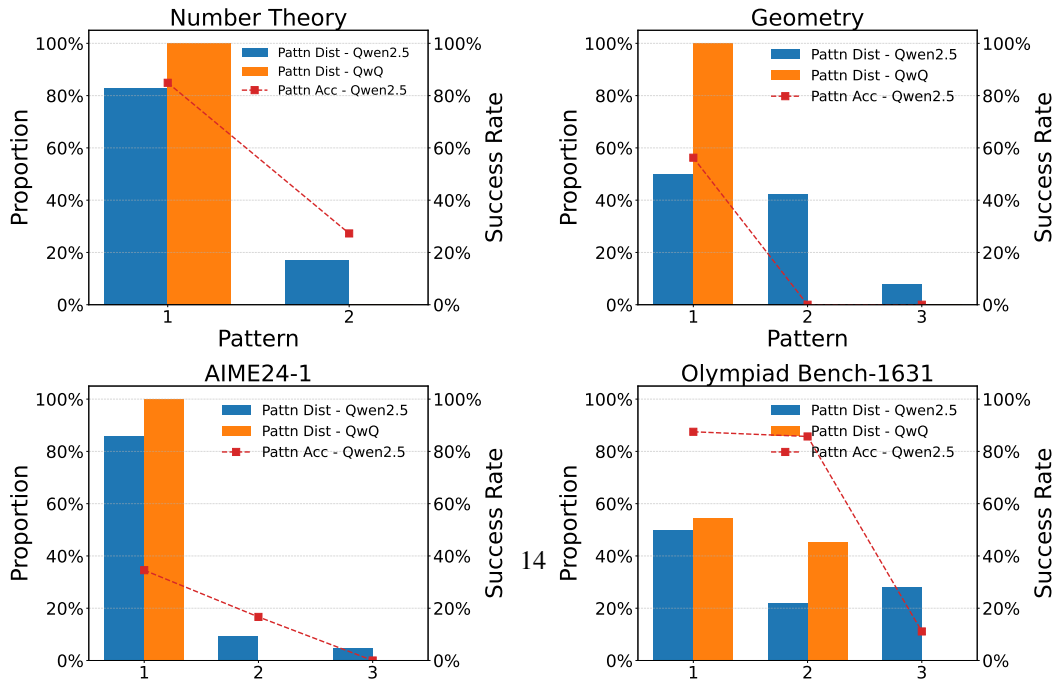


Figure 4: Additional reasoning-pattern evaluations with and without RLVR enhancement. Bars (Pattern Dist) show each pattern’s proportion, and dots (Pattern Acc) show the corresponding accuracy or success rate. Across tasks, RLVR shifts selection toward higher-success patterns, matching the paper’s empirical and theoretical conclusions.



B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: To support the main paper’s comparison between reward mechanisms, we use Qwen2.5-3B as the base model and train it on the MATH dataset with both RLVR and RLIF. We implement training with the verl [31] framework and keep all parameters identical to the MATH example provided by verl, changing only the reward calculation for RLIF. Different random seeds produce three training curves for each method, allowing the main-text trajectory comparison to reflect reward effects rather than configuration changes.

Token-level Analysis: This analysis supports the paper’s claim that sparse token changes can drive broader pattern-distribution shifts. We use Qwen2.5-3B and the RL-enhanced models described above. For GSM8K and MATH, we randomly sample 400 questions from each test set; for AIME24, we use the full dataset. For every question, we first sample an answer from the base model, concatenate that answer with the question, and use the result as a new prompt for the RL-enhanced model. We then use the OpenAI library interface to calculate each prompt token’s probability and its rank among all possible next tokens at that position. Comparing token ranks before and after RL gives the ranking-change ratio reported in the main text.

Reasoning-pattern Level Analysis: We use two complementary experiment types to support the pattern-selection contribution. First, we study pattern-distribution dynamics during training with Qwen2.5-3B and RL-enhanced Qwen2.5-3B, evaluating every 20 steps. Second, to test whether the conclusions extend to other base models and larger models, we compare Qwen-2.5-7B-SimpleRL-Zoo, QwQ-32B, and their corresponding base models. For these open-source checkpoints, we observe pattern distributions before and after RL rather than full training trajectories, because intermediate training checkpoints are unavailable.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: The inference stage is designed to collect enough responses to identify and track reasoning patterns. We randomly sample questions across task types such as Geometry and Algebra and across difficulty levels such as Level 4 and Level 5 from datasets including MATH. For pattern-distribution dynamics, each question receives 1024 sampled answers. For before-and-after RL comparisons, each question receives between 64 and 1024 sampled answers depending on the task category. These response sets are then used for pattern extraction and classification.

Reasoning Patterns Extraction: To construct fixed pattern categories, we sample a subset of base-model responses and use the GPT-4o [19] API to summarize recurring reasoning-pattern categories. The full extraction prompt appears in Appendix D.1. We set the GPT-4o API temperature to 0 so that the extraction procedure is deterministic and easier to audit.

Responses Classification: After defining the categories, we use GPT-4o to classify every response into the identified reasoning patterns, again with temperature set to 0. This step produces the pattern distributions that underlie the main empirical contribution. The full classification prompt appears in Appendix D.2.

Accuracy Analysis: To isolate the contribution of pattern distribution from changes in the reasoning process itself, we construct prompts that fix the reasoning prefix. Specifically, we concatenate the original prompt with a partial base-model response that represents a particular reasoning pattern. We remove the sentence containing the final answer, append the preceding sentence to the question, and ask the RL-enhanced model to continue predicting the answer. The accuracy of this continuation is treated as the success rate of the corresponding reasoning pattern, allowing us to test whether performance changes arise from pattern selection rather than pattern-specific improvement.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning-pattern examples used to classify responses across tasks.

B.3 Details for Each Task

This subsection documents the task procedures that support the additional pattern-selection experiments, including task descriptions and representative reasoning-pattern examples (Table 2). Example task prompts are provided in Appendix D.3.

Number Theory: The Number Theory task tests pattern selection in coin-distribution problems. Each problem starts with bags containing equal numbers of coins; after extra coins are added and redistributed equally, the total must exceed a specified value while preserving equal distribution. The model must determine the minimum initial coin count per bag. We derive the task from a level 5 MATH problem (id: test/number_theory/1055) [16] and create 32 variants by adjusting bag counts and coin totals. We use one-shot prompting with a simple unrelated example to enforce "boxed" answer formatting without steering the reasoning strategy. For evaluation, we select 20 questions with 4 responses each for pattern extraction, sample 64 responses per question, and classify all responses into the extracted patterns to compare pre- and post-RL distributions.

Geometry: The Geometry task tests whether pattern selection appears in triangle-inequality reasoning. We provide two side lengths of a scalene triangle and ask how many integer-centimeter values the third side can take. The task is derived from a level 4 MATH geometry problem (id: test/geometry/1046). We follow the Number Theory pipeline: construct a 32-question synthetic dataset, sample 20 questions with 4 base-model responses each from Qwen2.5-7b-Instruct for pattern extraction, and then compare models with and without RLVR enhancement on all evaluation questions with 64 responses each for pattern analysis.

606 **Algebra:** The Algebra task uses a can-recycling problem to elicit iterative reasoning patterns.
607 Starting from a fixed number of old cans, the model is told that n old cans can be recycled into one
608 new can and must determine the total number of cans eventually produced. The challenge is that each
609 recycling step may leave extra cans that can later combine with newly produced cans. The template
610 comes from a level 4 MATH algebra problem (id: test/algebra/2768). For this and the following
611 supplementary tasks, we use the same dataset-size and sampling settings as in the preceding tasks.

612 **Calculus:** The Calculus task provides a simple setting for pattern analysis in trigonometric equation
613 solving. We present a single-variable trigonometric equation, specify the variable’s domain, and ask
614 the model to count the number of solutions. Although the task is straightforward, it tests whether
615 RLVR still shifts selection among distinct solution patterns. The prototype is a level 4 MATH
616 pre-calculus problem (id: test/precalculus/1140).

617 **Counting and Probability:** The Counting and Probability task examines pattern selection in com-
618 binatorial probability reasoning. The problem involves white and black balls, a random draw of two
619 balls, and the probability that one drawn ball is black and the other is white. Given the number of balls
620 of one color, the model must compute the minimum number of balls of the other color. The task is de-
621 rived from a level 4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

622 **AMC23 Question:** For the AMC23 experiment, we use question 19 from AMC23 [1]. We
623 randomly sample 48 responses for reasoning-pattern extraction and then sample 1024 responses for
624 reasoning-pattern analysis.

625 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

626 For the QwQ-32B and Qwen-2.5-32B comparison, we run four task-level analyses under API
627 constraints. We randomly select one question per task, sample 64 responses from each model, and
628 perform reasoning-pattern extraction and classification directly from these answers. For MATH, we
629 use question 1055 from Number Theory and question 1046 from Geometry. For the other tasks, we
630 use problem 1 from AIME24 and question 1631 from OlympiadBench. Table 2 shows representative
631 pattern examples.

632 C Deferred Proofs

633 This appendix supports the paper’s theoretical contribution by giving the deferred derivations for the
634 optimal-policy characterization and the RLVR/RLIF training-dynamics results. We provide proofs
635 for Proposition 5.2, Theorem 5.3, Theorem 5.4, and Theorem 5.6.

636 C.1 Proof of Proposition 5.2

637 *Proof of Proposition 5.2.* We begin with the optimal-policy characterization, which formalizes how
638 the RL objective reweights reasoning-pattern probabilities. The proof follows techniques from [26].
639 Recall that the RL optimization objective is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

640 Under our framework from Section 5.1, the policy first samples $\mathbf{r}_i \in \mathcal{R}$ according to $\pi_{\theta}(\mathbf{r}_i | \mathbf{q})$ and
641 then produces the final answer $\mathbf{a} \in \mathcal{A}$ according to $\pi_{\theta}(\mathbf{a} | \mathbf{r}_i)$. With this decomposition, the reward
642 objective $\phi_{\text{RL}}(\theta)$ can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{r} | \mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

643 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r}) r_\phi([\mathbf{r}, \mathbf{a}], q)$, then

$$\begin{aligned}
\phi_{\text{RL}}(\boldsymbol{\theta}) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q}), \mathbf{a} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r})}{\pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r})} \right) \right] \\
&= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q})}{\pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})} \right) \right] \\
&= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q})} \left[\ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})} \right) - \ln Z \right] \\
&= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} \mid \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})] + \beta \ln Z.
\end{aligned}$$

644 Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})$ is the
645 partition constant that ensures

$$\pi^*(\mathbf{r} \mid \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q}),$$

646 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r} \mid \mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

647 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned}
&\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y} \mid \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} \mid \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x})] \\
&= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} \mid \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})] - \beta \ln Z.
\end{aligned}$$

648 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
649 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r} \mid \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

650 specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned}
R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i \mid \mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r})) \right) \\
&= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i \mid \mathbf{q})) + \log |\mathcal{A}| \right)}_c,
\end{aligned}$$

651 where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned}
\pi_{\text{opt}}(\mathbf{r} \mid \mathbf{q}) &= \frac{1}{Z} \exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q}) \\
&= \frac{\exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i)) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i \mid \mathbf{q})} \\
&= \frac{\exp \left(\frac{1}{\beta} (R'(\mathbf{r}) - c) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i \mid \mathbf{q})}
\end{aligned}$$

652 So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r}))$. This concludes the proof
653 of the theorem. $\square$

654 C.2 Dynamics of Tabular Policy

655 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
 656 gradient dynamics of θ can be computed as

$$\begin{aligned}\frac{d}{dt}\theta(t) &= \nabla\phi_{\text{RL}}(\theta(t)) \\ &= \nabla\mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla\pi_{\theta(t)}(\mathbf{r}|\mathbf{q})],\end{aligned}$$

657 Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
 658 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned}\frac{\partial\theta_{\mathbf{r}_i,\mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla\text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})\end{aligned}$$

659 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we
 660 have

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)} \cdot \frac{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)}{\partial t}$$

661 C.3 Proof of Theorem 5.3

662 To proof Theorem 5.3, here we first prove that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then we
 663 establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5), finally we derive the final bound for t stated in
 664 Theorem 5.3.

665 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial\theta_{\mathbf{r}_i,\mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

666 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned}\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)} \cdot \frac{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}),\end{aligned}$$

667 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \tag{C.4}
\end{aligned}$$

668 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
669 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
670 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

671 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
672 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

673 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

674 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

675 This concludes the proof of the theorem. □

676 C.4 Proof of Theorem 5.4

677 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}' | \mathbf{q})$ (Eq C.6), then we analyze
678 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

679 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 680 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

681 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

682 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

683 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 684 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

685 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

686 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

687 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

688 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

689 Let $C_2 = 1/\Delta, C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

690 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

691 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

692 This concludes the proof of the theorem. $\square$

693 C.5 Proof of Theorem 5.6

694 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
695 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, q))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$ where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, q)) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, q)) \right). \end{aligned}$$

As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$. By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated around the its expectation 0.5, the deviation is of order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$, the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$ converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$ is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables. Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) = 1 - 6\delta$.
- Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) = 1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

719 This concludes the proof of the second statement.

720

□

721 D Full Prompts

722 This appendix supports the empirical contribution by making the prompt templates for pattern
723 extraction, response classification, and task evaluation explicit. The prompts are reported verbatim so
724 that the reasoning-pattern pipeline can be inspected and reproduced.

725 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

726

Example processes from input
Common elements
...
There are reasoning processes:
{extracted_responses}
Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

727

728 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.
{patterns_categories}
Here are the reasoning processes to classify:
{extracted_responses}
Please output your classification in this JSON format:

729

730 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_endl>
<lim_start>
user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_endl>
<lim_start>assistant
Let me solve this step by step.
<think>

731

Prompt Example for Long Arithmetic

<lim_start>system
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_endl>
<lim_start>user
Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_endl>
<lim_start>assistant
Let me solve this step by step.
<think>

732

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

733

Prompt Example for Geometry

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

734

Prompt Example for Algebra

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

735

Prompt Example for Calculus

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

736

Prompt Example for Counting and Probability

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

737

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of ' $8^5 \cdot 5^{10} \cdot 15^5$ '?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

738

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a 9 -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

739

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

740